2024 BUDDING VALUE INVESTOR AWARD

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Part 1

The miracle rise of China as one of the high-growth nations in the world over the last few decades has been fascinating. Nonetheless, as the current trend rolls out to be unfavorable to the Chinese economy, the nation has been struggling. China is facing the deglobalization trend or, in other words, countries are diversifying away from China, stemming from the impact of Covid-19 pandemic. The misfortune does not end there, the Chinese property market collapsed in 2021 following the bankruptcy of Evergrande Group. A series of unfortunate events have sent the Chinese stock market into a free fall to extremely low prices, evoking comparisons to Japan's Lost Decades. Two polarized stories have since emerged with one suggesting that China's stock market is uninvestable or presents a value trap while the other remains optimistic about the upside potential given the hugely discounted price-to-earnings ratio. This article aligns with the latter perspective, as Figure 1 shows that the Chinese companies are not experiencing a significant decline in performance. In fact, they manage to generate desirable free cash flow while maintaining the profit margin amidst the challenging market conditions. Chinese companies are recording revenue and net income growth with minimal debt dependency, indicating efficient asset utilization. Most importantly, they are generating significant free cash flow, enabling them to return value to shareholders or reinvest for further growth.

Search		Net margin TTM	ROIC TTM	Revenue growth Annual YoY	Net income growth Annual YoY	Debt / assets FY	FCF FY
601398	INDUSTRIAL AND COMMERCIAL BANK OF CHINA LIMITED D	—	—	+7.89%	+0.79%	0.14	499.189 B CNY
600519	KWEICHOW MOUTAI CO.,LTD. D	58.34%	38.30%	+17.67%	+19.16%	0.05	64.114 B CNY
601288	AGRICULTURAL BANK OF CHINA LIMITED D	—	—	+11.16%	+3.91%	0.20	310.725 B CNY
601857	PETROCHINA COMPANY LIMITED D	—	—	−8.32%	+8.34%	0.16	174.296 B CNY
600941	CHINA MOBILE LTD D	13.24%	9.52%	+7.69%	+5.03%	0.05	134.473 B CNY
601988	BANK OF CHINA LIMITED D	19.13%	6.97%	+19.75%	+2.38%	0.19	514.759 B CNY
601939	CHINA CONSTRUCTION BANK CORPORATION D	—	—	+5.80%	+2.44%	0.17	560.6 B CNY
600938	CNOOC LIMITED D	30.42%	16.85%	+2.28%	−12.60%	0.12	92.322 B CNY
600036	CHINA MERCHANTS BANK D	29.09%	6.77%	+3.17%	+6.39%	0.14	181.393 B CNY
600028	CHINA PETROLEUM & CHEMICAL CORPORATION D	—	—	−3.76%	−9.87%	0.22	9.797 B CNY
300750	CONTEMPORARY AMPER D	12.29%	16.14%	+21.83%	+43.58%	0.28	60.913 B CNY
601088	CHINA SHENHUA ENERGY COMPANY LIMITED D	—	—	+0.04%	−14.29%	0.06	57.227 B CNY
601628	CHINA LIFE INSURANCE COMPANY LIMITED D	4.36%	4.75%	−7.00%	−30.74%	0.05	551.471 B CNY
600900	CHINA YANGTZE POWER CO.,LTD. D	35.91%	6.81%	+13.51%	+14.81%	0.55	54.637 B CNY

Figure 1: List of Chinese public companies

If the Chinese stock issues are not with the companies themselves, then what are they?

Hit hard by the Covid-19 outbreak, compounded by an aggressive zero-Covid policy, followed by a property market bust, the Chinese economy has entered a downward spiral. These challenges have significantly undermined both consumer and investor confidence. The property market, which contributes 20% of fiscal revenue, 24% of the gross domestic product (GDP) and 70% of household wealth, has been a key driver to the Chinese economy (Liu, 2024). Past decades of rapid urbanization fueled the development of new homes and buildings, strengthened by the cultural emphasis on property ownership. However, it failed to sustain prosperity with excessive credit and regulatory failure. With such a significant portion of wealth being wiped off, household confidence and purchasing power have nosedived together with the property value. Besides, Chinese culture views property as one that provides a sense of belonging, stability and rooting for a family; one that symbolizes wealth and status. Having been defeated by an important pursuit of their lives, one could hardly pick up the appetite to invest, at least soon.

This is not the first time investor confidence has taken a hit. The Chinese stock market experienced a boom in 2014-2015, fueled by the government communication campaign to encourage citizens to participate in the market. It saw over 38 million new investment accounts opened in just two months. When the bubble popped, the inexperienced retail investors were left panicking and miserable (Best, 2023). The frequent undesirable experiences with the stock market could have a long-term traumatized impact on the domestic retail investors, leading to them being reluctant to invest.

Table 1 Income and Expenditure Nationwide in 2022

Item	Absolute Value	Y/Y(%) Actual Growth in Brackets
Total Per Capita Income Nationwide	**36883**	**5.0（2.9）**
Grouped by Permanent Residence		
Urban Households	49283	3.9（1.9）
Rural Households	20133	6.3（4.2）
Grouped by Income Source		
Income of Wages and Salaries	20590	4.9
Net Business Income	6175	4.8
Net Income from Property	3227	4.9
Net Income from Transfer	6892	5.5
Total Per Capita Income Nationwide Median	**31370**	**4.7**
Grouped by Permanent Residence		
Urban Households	45123	3.7
Rural Households	17734	4.9
Total Per Capita Expenditure Nationwide	**24538**	**1.8（-0.2）**
Grouped by Permanent Residence		
Urban Households	30391	0.3（-1.7）
Rural Households	16632	4.5（2.5）
Grouped by Consumption Category		
Food, tobacco and liquor	7481	4.2
Clothing	1365	-3.8
Residence	5882	4.3
Household facilities, articles and services	1432	0.6
Transportation and telecommunication	3195	1.2
Education, culture and recreation	2469	-5.0
Health care and medical services	2120	0.2
Miscellaneous goods and services	595	4.6

Figure 2: NBS Households' Income and Consumption Expenditure in 2022

Nationwide Income and Expenditure in 2023

Item	Absolute Value	Y/Y(%) Real Growth in Brackets
Nationwide per Capita Disposable Income	39218	6.3（6.1）
Grouped by Permanent Residence		
Urban residents	51821	5.1（4.8）
Rural residents	21691	7.7（7.6）
Grouped by Income Source		
Income from wages and salaries	22053	7.1
Net business income	6542	6.0
Net income from properties	3362	4.2
Net income from transfer	7261	5.4
Median of the nationwide per capita disposable income	33036	5.3
Grouped by Permanent Residence		
Urban residents	47122	4.4
Rural residents	18748	5.7
Nationwide per Capita Consumption Expenditure	26796	9.2（9.0）
Grouped by Permanent Residence		
Urban residents	32994	8.6（8.3）
Rural residents	18175	9.3（9.2）
Grouped by Consumption Category		
Food, tobacco and liquor	7983	6.7
Clothing	1479	8.4
Residence	6095	3.6
Household facilities, articles and services	1526	6.6
Transportation and telecommunication	3652	14.3
Education, culture and recreation	2904	17.6
Health care and medical services	2460	16.0
Miscellaneous goods and service	697	17.1

Figure 3: NBS Households' Income and Consumption Expenditure in 2023

The Chinese population may shun away from investment, but I disagree with the notion that Chinese spending remains gloomy. Instead of saying they consume less, it would be more accurate to say that their spending patterns have changed from material goods to experiences. Look at Figure 2 and 3, 2023 saw a 9% real growth in per capita consumption after recording a decline in 2022. Consumption has picked up in all categories with some even registering double-digit growth, including transportation and telecommunication; education, culture and recreation; healthcare and medical services; as well as miscellaneous goods and services. Contrary to the gloomy comments on the Chinese consumption trend, people have indeed been spending but they now seek satisfaction more from experiential consumption.

Figure 4: China's Real GDP Growth ©IMF, 2024, Source: World Economic Outlook (April 2024)

Although China's GDP growth is slowing down, it is not undesirable with a 5.2% real GDP growth in 2023. Years of double-digit growth experienced by the nation have sparked interests and heated discussion from the outsiders about its fullest potential which may have associated China with an unrealistic expectation. As the world is watching the end of China's cheap manufacturing powerhouse era, the country is transitioning to stabilize their growth rather than pursuing a consistently high level of miraculous expansion while maintaining its competitive position, supported by its strong GDP performance. To put it into context, China accounts for 19.01% of global GDP based on purchasing power parity, surpassing the United States and showing gradual growth. Looking at a different measure, a similar upward trend is observed in the GDP per capita, in terms of current price. Despite challenges, the figures demonstrate China's strength in generating economic value and prosperity.

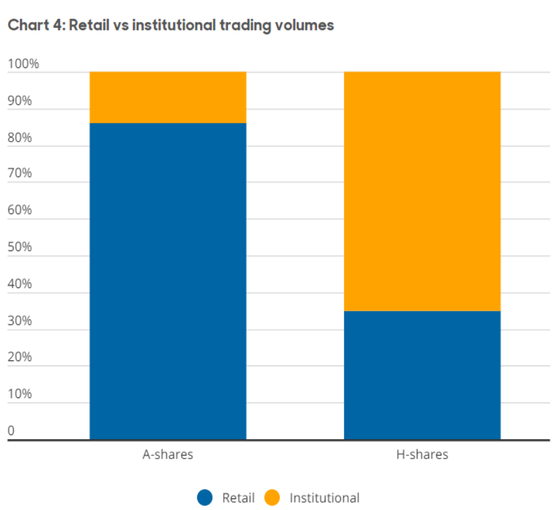

Figure 5: Retail vs institutional trading volumes

A question may then arise, why doesn't the Chinese stock market seem to reflect the improvement in the economic condition. This frustrating phenomenon could be attributed to several factors. As depicted by Figure 5, local retail investors account for up to 86% of China A-shares trading volumes, compared to a mere 14% from institutional investors. On the contrary, institutional investors account for approximately 80% of the volume of trades on the New York Stock Exchange (Palmer, 2023). Consequently, China's stock market is much more susceptible to emotional trading and excessive volatility driven by speculation without the backing of concrete research. The instability of the market is evident by the 2015 stock market bubble burst where it lacks the huge funds from the institutional investors to strengthen the foundation.

The imbalance proportion has also amplified the negative effect from the extremely weak investors' confidence owing to the property market slump. In addition, Chinese are well known for their high propensity to save. According to Lee (2024), household net bank deposits have surged to over CNY 30 trillion, equivalent to over 20% of China GDP over the past three years. Zisper (2023) echoes the same findings where there are RMB 53 trillion in savings accumulated since 2020 which equates to an extra RMB 38,000 in each consumer's bank account. Having a conservative culture of savings and now deepened by the diminished confidence, the Chinese population is not putting money anywhere. Since they contribute to the most trading volume in the A-share market, there is insufficient inflow to boost the stock value upwards even if the Chinese companies are reporting desirable performance.

On top of that, there is a mismatch between China's GDP composition and its market index. For instance, the financial sector weighs 33.66% of the Hang Seng Index (HSI) and 29.71% in information technology. Similarly, 6 of the top 10 companies on the Shanghai Stock Exchange are categorized under the financial sector (FSMOne, 2023). However, this sector contributes only 8% to GDP in 2023, falling far behind the industrial sector that distributed 31.7% according to Statista as shown in Figure 7. Nonetheless, the industrial sector has a weighting of only 0.76% in the HSI.

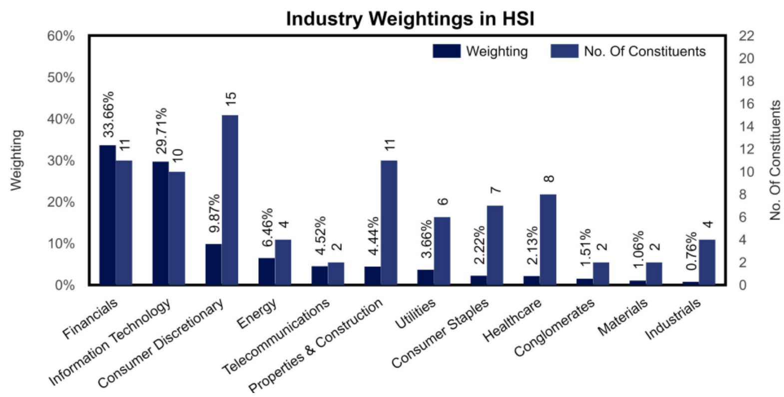

Figure 6: Industry weightings in HSI

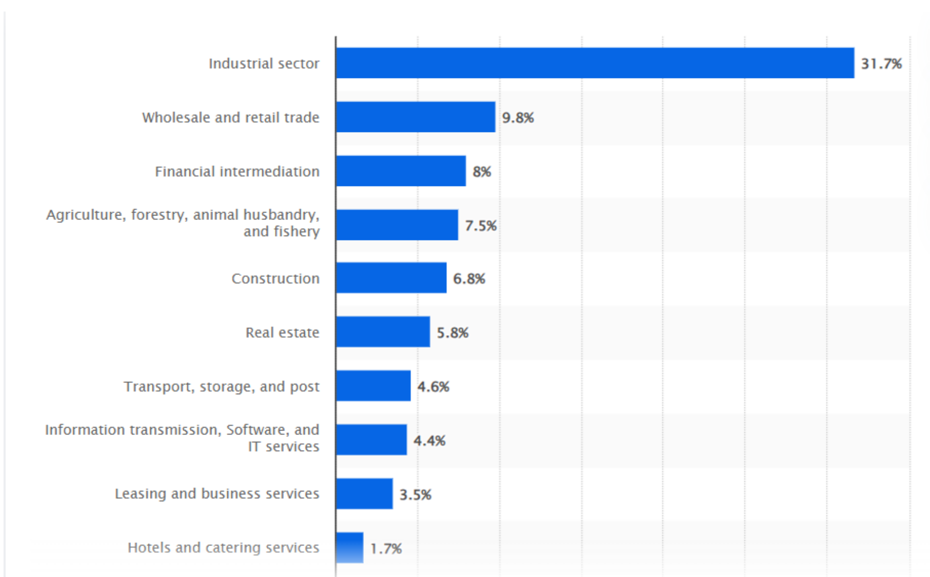

Figure 7: Distribution of the gross domestic product in China in 2023 by industry

The overconcentration in stable, traditional sectors like finance offers limited growth prospects. Moreover, the mismatch leads to the failure to capture or accurately reflect the performance of high growth companies on the indices. Index oftentimes acts as a mirror to reflect the overall capital market performance of a nation, hence, people started to perceive that China's market is uninvestable. The heavy focus on the financial sector, especially after the property market collapse, further exacerbates the underperformance of the Chinese stock market. Referring to Figure 8, housing loans dominate the retail loan structure of many banks, leading to poor loan performance that dragged down the broader economy due to the market's concentrated structure.

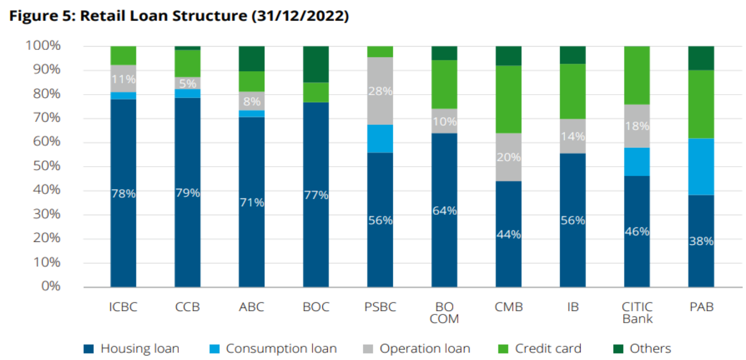

Figure 8: Chinese banking sector 2022 review and 2023

In recent years, the Chinese government has attempted to stimulate the domestic market by lowering market interest rate or prime rate. However, the mismatch of the market structure seems to have offset the positive effect of an expansionary policy as the interest margin of the banks shrink as shown in Figure 9. The banking net interest margin has dropped to a record low in 2022. When the deterioration reflects on the capital market and indices, how would the consumers or investors feel the confidence to spend and invest?

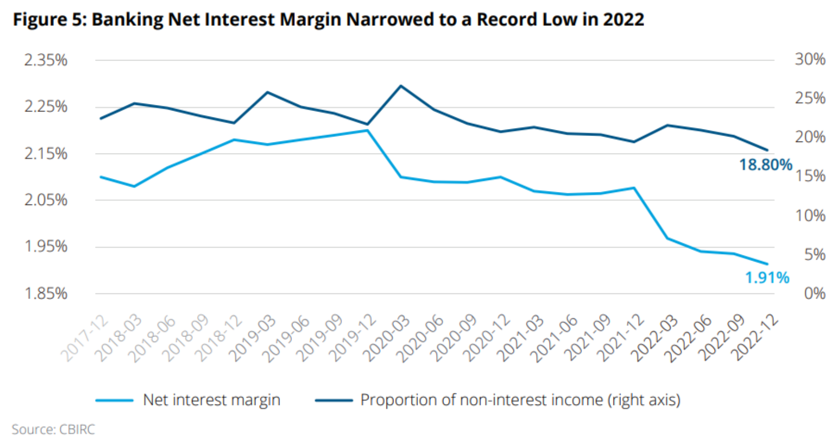

Figure 9: Chinese banking sector 2022 review and 2023 outlook

As previously stated, China may no longer benefit from its once internationally competitive manufacturing capacity, due to the deglobalization trend and the need for structural change if the nation aims to become a highly developed country. As resources shift to other high-growth areas, investors may face a prolonged consolidation period due to uncertainties during the transition or reform. According to a book published by the IMF, China is shifting its economy from external to domestic demand; industry to services; state-led to more market-driven resource allocation (Lam et al., 2017). While evolution is necessary for growth, which invites unknowns, doubts and sometimes periods of darkness, the long-term results can be rewarding. In short, while risks persist in the Chinese market, they stem more from the macroeconomic sentiment and market structural or policy issues than from poor Chinese corporate performance.

Part 2

Company Overview

Contemporary Amperex Technology (CATL), founded in 2011 in Ningde, China, is a leading manufacturer specializing in electric vehicle (EV) batteries, energy storage systems and battery material recycling. Led by the co-founder Zeng Yuqun, CATL has expanded its operations across 11 cities of China, with additional production sites in Germany and Hungary. Apart from production, CATL has 6 research and development centers with 5 of them based in China and the other one in Munich, Germany.

Serving primarily in the business-to-business (B2B) space, CATL provides battery solutions for passenger vehicles, commercial applications, energy storage systems and battery recycling. After establishing a solid business foundation, CATL has since listed on the Shenzhen Stock Exchange in 2018 with a stock code of 300750. Its current stock price is CNY 182.90 as of September 10, 2024. The company operates with a vision of: 'Rooted in Chinese civilization and embracing global culture, striving to be a world-class technology innovator, delivering superior contributions to green energy for the world, and providing a platform of pursuing spiritual and material well-being for employees!'. The battery manufacturer focuses its growth strategy in 3 key areas: replacing stationary fossil energy with renewable energy generation and energy storage; replacing mobile fossil energy with EV batteries; realizing integrated innovation of market applications with electrification and intelligentization.

Industry Outlook

Amidst the transformative forces of a rapidly evolving world, one industry stands as a beacon of progress - the battery industry. With the important role played by batteries in decarbonizing transport, enabling the shift to renewable power generation, and providing access to electricity to off-grid communities, batteries have become the heartbeat of the modern world (World Economic Forum, 2019). As the world is phasing out internal combustion vehicles that rely on fossil fuel, batteries are used as the main energy source to power human mobility. As illustrated in Figure 10, the first quarter of 2024 recorded strong electric vehicles sales, exceeding the same quarter in 2023 by 25%. Today, the Chinese population is the main contributor to the global EV sales due to their early pivot to the clean energy field, accounting for approximately 60% of new electric car registration in 2023. In contrast, the markets in the US and Europe are far less saturated, presenting a great upside potential with supportive government

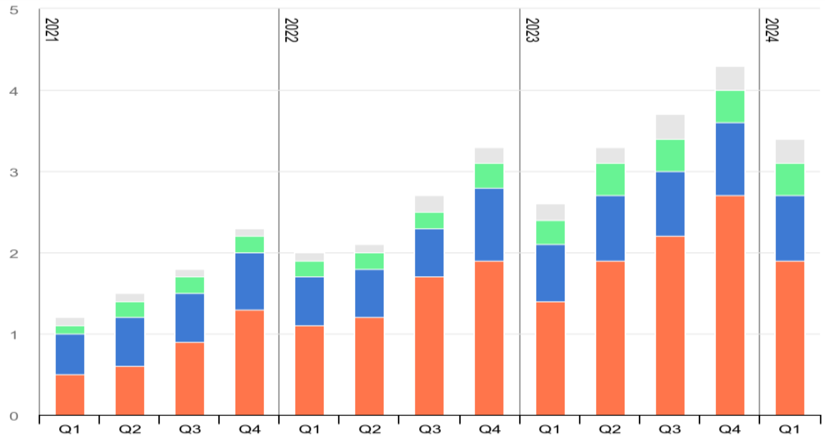

Figure 10: Quarterly electric car sales by region, 2021-2024

policies. For instance, the sales of the Tesla Model Y increased 50% in 2023, after it became eligible for the full USD 7,500 tax credit under the Clean Vehicle Tax Credit (Global EV Outlook 2024: Moving towards increased affordability, 2024).

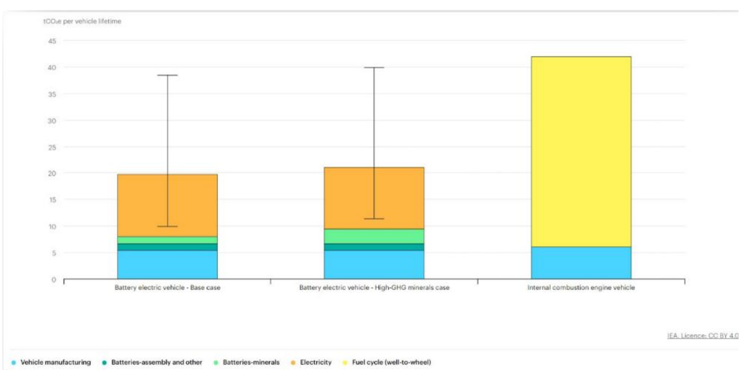

Figure 11: Comparative life-cycle greenhouse gas emissions of a mid-size BEV

However, it is insufficient to achieve the net zero target by merely switching to electric vehicles. As suggested by Figure 11, the lifetime emission of a battery electric vehicle comes mainly from electricity. What is powering the EV charging station needs transformation as well. Due to the intermittent nature of renewable energy sources, power grids still depend on the burning of fossil fuel. Here then arises another pivotal application for batteries: renewable energy storage systems. These systems, often involving batteries with substantial storage capacities, play an indispensable role in addressing the intermittent nature of renewable energy sources. They store excess energy generated during peak production periods and release it during times of low production or high demand. This mechanism yields a more consistent energy output, effectively curbing the reliance on fossil fuel-driven power sources. Furthermore, in the context of critical scenarios and power disruptions, the integration of batteries within uninterruptible power supply (UPS) systems functions as steadfast backup power reservoirs. In fact, more than $5 billion was invested in battery energy storage systems (BESS) in 2022, almost a threefold increase from the previous year. The market is expected to reach between $120 billion and $150 billion by 2030, more than double its size today (Jarbratt et al., 2023).

A milestone has been achieved in 2023 where renewable energy accounted for more than 30% of the world's electricity for the first time (Ambrose, 2024). This shows the world's dedication to eliminate the use of fossil fuel and to smoothen the transition, as such, BESS will face substantial demands. According to the International Energy Agency, total volume of batteries used in the energy sector was over 2,400 gigawatt-hours (GWh) in 2023, a fourfold increase from 2020. In the past five years, over 2,000 GWh of lithium-ion battery capacity has been added worldwide, powering 40 million electric vehicles and thousands of battery storage projects (IEA, 2024). Moreover, the International Energy Agency (IEA) projects a surge in global EV battery demand under its "Stated Policy Scenario", quadrupling to 3 terawatt hours (TWh) by 2030, fueled by increasing worldwide EV sales and a concurrent uptrend in per-vehicle battery capacity. They also foresee a compounded annual growth rate of over 20% in global energy storage installations up to 2030, supported by robust policy initiatives (Chao, 2024). Evidently, the global movement towards clean energy is bringing a huge tide to the battery industry.

National Policy Support

The demand is further fueled by the support from national policies and international organizations. In 2016, the Paris Agreement was signed, committing 195 Parties to the need for an effective and progressive response to the urgent threat of climate change on the basis of the best available scientific knowledge. Countries are expected to submit Nationally Determined Contribution, an updated action plan to tackle emission issues every five years (United Nations, n.d.). Furthermore, an agreement was made during the UN's Cop28 climate change conference to grow renewables to 60% of global electricity by 2030 (Ambrose, 2024). Publicly delivering such a goal or promise, countries are held accountable in their actions to achieving environmental sustainability.

Besides, the introduction of the Inflation Reduction Act signifies the determination of the US government in tackling environmental challenges and advancing its competitiveness in the clean energy sector. Nearly $400 billion has been allocated to clean energy investments, with clean electricity and transmission taking up the largest slice, followed by clean transportation. The Act goes to great lengths to incentivize private corporation investments in clean energy, transportation and manufacturing with $216 billion worth of tax credits, thereby leveraging on the influences American corporations have on a global scale (Badlam et al., 2022).

European countries are pursuing a similar path in contributing to global efforts. Policies are enacted to push for more aggressive actions and initiatives, for example, the Net Zero Industry Act sets a goal to satisfy 40% of net-zero technology needs via local manufacturing by 2030, including batteries (European Commission, 2023). The plan aims to simplify permitting for facilities like battery production, offering quicker access to subsidies and loans to address energy costs and liquidity as well as prioritizing worker reskilling and trade resilience. Clearly articulated by their actions, the world is ramping up to save the planet, making the motherland livable and sustainable, which directly gives rise to the clean energy industry. To put it into context, global investment in EV batteries has surged eightfold since 2018 and fivefold for battery storage, rising to a total of USD 150 billion in 2023 (IEA, 2024).

China’s domination in the clean energy sector

China is one of the earliest adopters of clean energy which has paved for its technological advancement in today’s economy. They had taken steps to include EV technology as a priority research project in China’s Five-Year Plan and invested in related technologies back in 2001 (Yang, 2023). With decades of knowledge and expertise, China has built a highly competitive clean energy industry in the nation, supplying the Chinese technologies to the rest of the world. To highlight the significance, China dominates the battery supply chain with nearly 85% of global battery cell production capacity and substantial shares in cathode and anode active material production (IEA, 2024). According to the projection made by Statista in Figure 12, China is expected to hold 65.2% of lithium-ion battery production capacity in 2025, far outweighing the foreign nations even after the estimated decline from 79% in 2021.

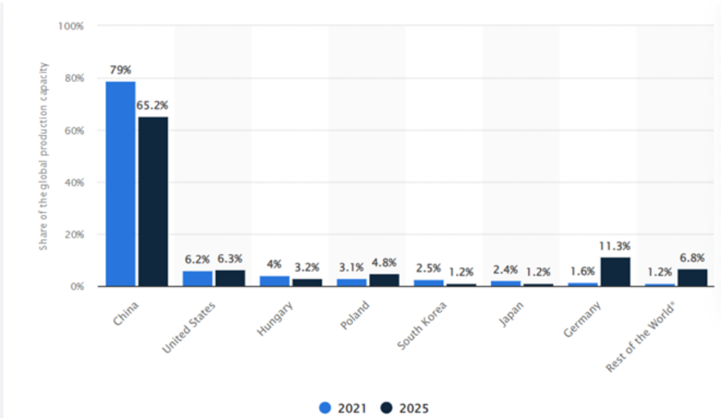

Figure 12: Share of the global electric vehicles lithium-ion battery manufacturing capacity in 2021 with a forecast for 2025, by country.

The success of China's clean energy sector can be attributed to the substantial demand for electric vehicles and energy storage systems. As mentioned above, most EV sales come from the Chinese population. Besides, the application of energy storage systems is expected to increase as China is committed to lowering its dependency on fossil fuel, which still accounts for 70% of electricity generation (Hilton, 2024). In 2022, China installed nearly as much solar photovoltaic capacity as the rest of the world combined. In 2023, the nation doubled the new solar installation, increased 66% of new wind capacity and almost quadrupled its energy storage additions (Hilton, 2024).

In essence, multiple factors such as the soaring international demands, supportive policies and substantial investments in the clean energy sector have developed a favorable ecosystem for the Chinese companies to thrive.

Competitive Advantage

Huge market share

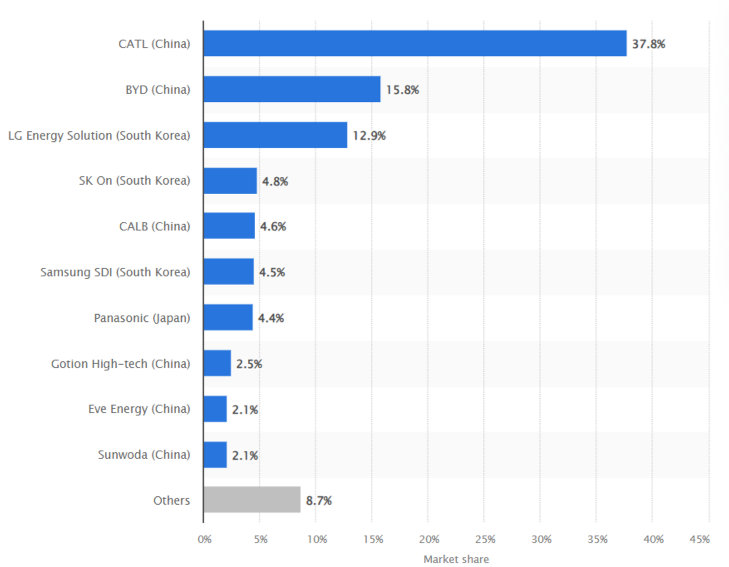

Figure 13: Global market distribution of battery makers for electric vehicles in 2023

One of the competitive advantages of CATL is its significant market share of 37.8%, as shown in Figure 13. The Chinese company BYD ranked second with a market share of 15.8%, followed by LG Energy Solution with 12.9%. There is a huge market gap between the first and second place, which means CATL has a far greater outreach and influence on the battery industry (Global market distribution of battery makers for electric vehicles in 2023, n.d.). According to The Wall Street Journal citing SNE Research, one in three EV sold worldwide used a lithium-ion battery produced by CATL (Thielen, 2024). CATL's extensive market coverage reflects strong customer confidence and satisfaction with its battery solutions. This builds strong brand recognition and trust that induces new customers' adoption and reinforces existing customer loyalty. Consequently, CATL creates a barrier for new entrants and limits the growth potential of existing rivals.

Operating on a larger scale, CATL could spread out its fixed costs, achieving economies of scale which could relieve the burden on operational expenses and retain desirable profit margin. Besides realizing cost reduction internally, large companies often have greater bargaining power with suppliers to negotiate for better terms which strengthen CATL's cost competitiveness. On top of that, with a larger market share, it comes with higher sales volumes, generating higher cash flows that could be reinvested in R&D to further reinforce its technological edge.

Huge International Customer Base

Strategically, CATL is expanding overseas to compensate for the slowing growth in the relatively saturated domestic market. According to SNE Research, CATL's market share in electromotive vehicles battery installations outside China rose sharply to 27.7% year-on-year in 8M23. The high overseas adoption rate contributed to the significant uptick of 70.29% year-on-year in the overseas revenue of the battery manufacturer to 130.992 billion RMB (Fitch Ratings, 2023). Overseas market has now accounted for 32.7% of CATL total sales in 2023, an approximate 9 percentage points increase from 2022. Impressively, CATL has achieved strong international sales growth despite the geopolitical trade tension. This reflects the widespread acceptance and competitive edge of CATL's products over its competitors.

Over the years, CATL has secured partnership with multiple pioneering car manufacturers such as BMW, Stellantis, Ford, Honda, Volvo and Hyundai, supplying mobile battery solutions and the necessary technologies for green transition. In the field of energy storage, Rolls-Royce and CATL signed a long-term supply agreement for more than 10 gigawatt-hours of storage capacity last year. Roll-Royce has, as well, launched CATL's TENER product line in the European Union and the United Kingdom (Liu, 2024). What's more, CATL has forged partnerships with significant national electricity suppliers such as Synergy in Australia and ENEL in Italy in assisting their energy storage projects. With growing reliance on CATL globally, the company is critical in supporting the domestic electricity supply with its collaboration with, for example, Huaneng Group and China Huadian Corporation.

Rapid market uptake and extensive customer base have enabled CATL to rank top globally in battery energy storage system battery shipment for three consecutive years as of 2023. Looking at a broader picture, usage volume of CATL batteries has ranked first globally for seven consecutive years. Once again, CATL has reinforced its leadership position in the battery industry, establishing a high barrier through its extensive market reach and influence which contribute to a strong economic moat.

Technology advancement

The wide market penetration and market share CATL enjoys should be attributed to the company's advanced technologies. With strong profitability and a huge talent pool of up to 20,604 R&D personnels, CATL can afford continuous innovations and experiments to solve existing limitations with batteries and satisfy customers' needs at a competitive level. CATL understands the market frustration and has successfully introduced various breakthroughs in the aspects of battery range, thermal control and carbon emissions. For example, CATL launched the world's first solar-plus-storage integrated solution with zero auxiliary power supply, getting rid of the reliance on cooling systems and auxiliary power supply through the self-developed solar-BESS converter, the heat-resistant cell technology and advanced self-heating technology (CATL, 2023). Recently, Shenxing PLUS was introduced as the world's first LFP battery that achieves a range above 1000 kilometers with 4C superfast charging (CATL, 2024).

To deliver high-level breakthroughs on a consistent basis, CATL acknowledges the importance of efficiency in the factory or lab. CATL received the Global Lighthouse Factory designation award in 2021 by the World Economic Forum, recognizing manufacturers who have applied advanced technologies such as artificial intelligence and big data analytics to achieve operational efficiency and profitable growth in tandem with environmental stewardship (Yan, 2022). To put it into context, CATL boasts the highest utilized capacity among its rivals as shown in Figure 14. Being at the front line of green transition, CATL has achieved a significant internal milestone where its Yibin Plant was certified as the world's first zero-carbon battery factory in 2022.

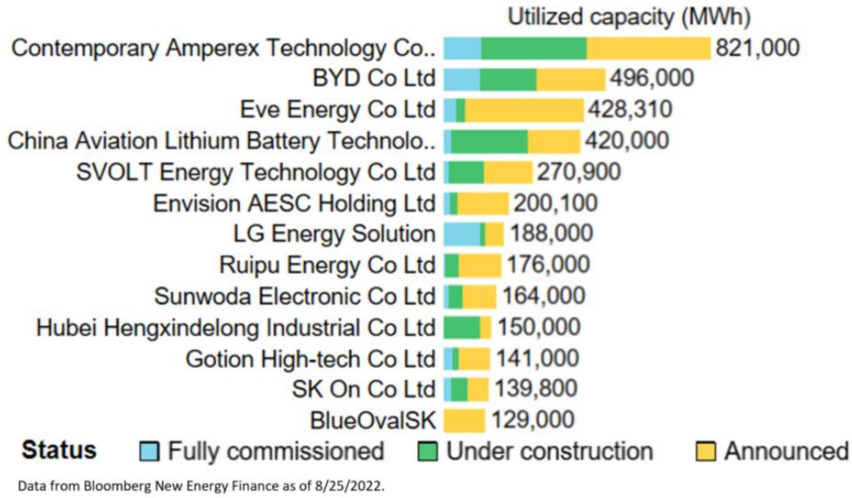

Figure 14: Rank by manufacturer (Utilized capacity, MWh)

Committing to deliver world-class battery solutions, CATL has positioned itself as a leader in technological innovation. The superior battery performance and reliability have enhanced their higher perceived value, fortifying its economic moat. This has further reinforced customer trust and loyalty, making it difficult for competitors to match its offerings. By setting a high industry standard, it solidifies its position as a trusted battery provider in the market.

Risk

Geopolitical tension

As a Chinese company operating on an international scale, CATL has no choice but to bear the challenges of geopolitical tension. Previously mentioned, Western countries, especially the US and Europe are seeking alternatives to replace Chinese goods with the concerns of national security as the two largest economies are fighting for the world leadership position. Although Chinese companies have, multiple times, denied such allegations, it could not seem to alleviate their fear. The US and Europe have shifted their efforts to bring clean energy technologies onshore with policies such as the aforementioned Inflation Reduction Act incentivizing the setting up of factories on the US soil. What is even worse, they take it a step further by imposing tariffs that target imports of EV, lithium-ion batteries and critical minerals from mainland China. Tariffs on batteries and battery parts will increase from 7.5% to 25%, taking effect on the 1st of August 2024 (Gogoi, 2024).

Inevitably, CATL will feel the pain as its customers may look for cheaper alternatives following the price hike from the import tariffs. This could hinder CATL's ability to expand internationally and sustain revenue growth, especially as the domestic market shows signs of saturation and the growth rate decelerates. It is even harder to navigate the global market when the US lawmakers perceive CATL itself has a close relationship with the Chinese Communist Party and its military which could endanger US national security (Reuters, 2024). The hostility towards China can be seen in the intervention of the US House of Representatives into the partnership between Ford Motor and CATL, in which Ford Motor was demanded to turn over the partnership documents and threatened to summon CEO Jim Farley to testify before Congress (Shepardson, 2023).

Such barriers could severely detriment trade efficiency and incur additional expenses to the business, making US companies to be hesitant or put an end to the trade relationship with CATL. However, the aggressive actions from the US could be unrealistic and could backfire as their clean energy technology is still years behind according to Michael Dunne, the founder of Dunne Insights. Dunne says it would take between five and 10 years for the US to catch up with China (Hawkins, 2024). As discussed in the previous section, China has a significantly larger production capacity than the US, and both countries are undergoing a clean energy transition that demands substantial green technologies. If the US could not provide similarly competitive technologies, the companies would be compelled to remain reliant on CATL, despite the additional costs. Notably, Tesla is planning to set up a battery factory in Nevada using CATL machinery this year (Hawkins, 2024). The technological advancement of CATL could be too critical for US companies to forgo.

Raw material price fluctuation

The production of batteries involves several critical raw materials such as lithium, nickel and cobalt. Unfortunately, the materials possess a nature of extreme price volatility which, in return, affects the gross margin of battery manufacturers. Prices of key battery metals – especially lithium have fallen dramatically since January, due to significant growth in production capacity across all parts of the battery value chain, from raw materials and components to battery cells and packs. What follows is the price cut of the lithium-ion battery with the average pack prices falling to US$139 per kilowatt hour (kWh) this year, a 14% drop from US$161 per kWh in 2022, according to the Bloomberg NEF's (BNEF) annual lithium-ion battery price survey (The Star, 2023). As observed in Figure 15, the value of lithium futures has been falling since the beginning of the year 2023. As quoted in CATL 2023 annual financial report, the price of lithium carbonate experienced a huge volatile movement, resulting in instability of the costs which could impact the cash flow of the company.

Figure 15: Lithium futures

Intensified competition

Following the rapid growth of the clean energy industry, new entrants are coming in to compete for market share. The growing concern comes, primarily, from the geopolitical tension as the world is undergoing a major reform. Countries are now adopting the ‘China + 1’ movement to diversify away from the Chinese suppliers, after having experienced the supply disruption during the Covid-19 pandemic and influenced by the US aggression against the world's second largest economy. Companies no longer partner with one supplier but many which would encroach the global market share of CATL. The intensified competition from its closest rivals such as BYD, Panasonic and LG Energy Solution require CATL to pour substantial funds into R&D activities consistently to maintain the technological edge it boasts, and the economic moat developed by the advanced battery production techniques it possesses. Such efforts could burden its long-term cash flows.

Nonetheless, the market share of CATL is less likely to be surpassed in the short term due to its advancement in technological capacity as well as its ample cash holding that can continuously finance R&D at a competitive level.

Financial Analysis

Common Size Analysis

CATL has a strong cash holding with cash and cash equivalent accounting for the largest share of assets at 36.85%. In 2023, it has a total value of CNY 264.31 billion, a 1550% increase from 2017. Net property, plant and equipment comes in second with a weightage of 20.76%. The third largest portion is total receivable at 18.5%. Receivables recorded a sharp rise in 2022 from CNY 40.37 billion to CNY 101.26 billion, following the significant revenue growth in 2022.

Looking at the components of liability, account payable accounts for 24.6% and holds the largest share. It reflects the dependency of CATL on credit terms to acquire necessary materials for battery production and sales. It could be beneficial to the cash flow in the short term as no significant initial outlay is necessary to service its customers, enabling them to direct the funds for promotional events. However, it is susceptible to external shocks whereby the battery demand will be influenced and put a toll on the sales performance or excessive high credit sales that delay the cash collection. Under these scenarios, CATL may not receive sufficient cash to meet its credit terms. Ranking second, long-term debt takes up 21.96% of total liability, followed by 20.95% of short-term debt.

Profitability Analysis

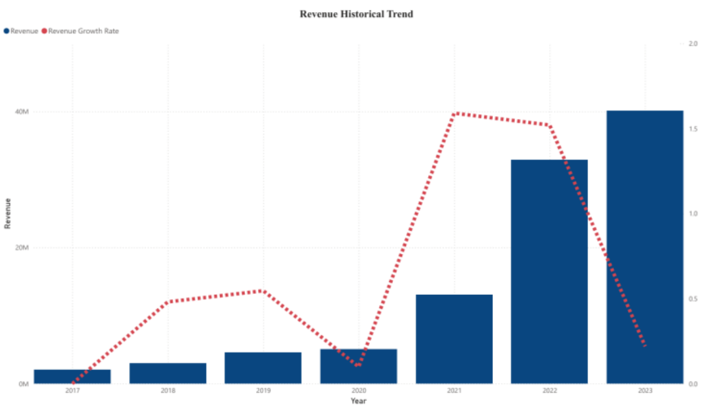

Figure 16: CATL’s revenue historical trend

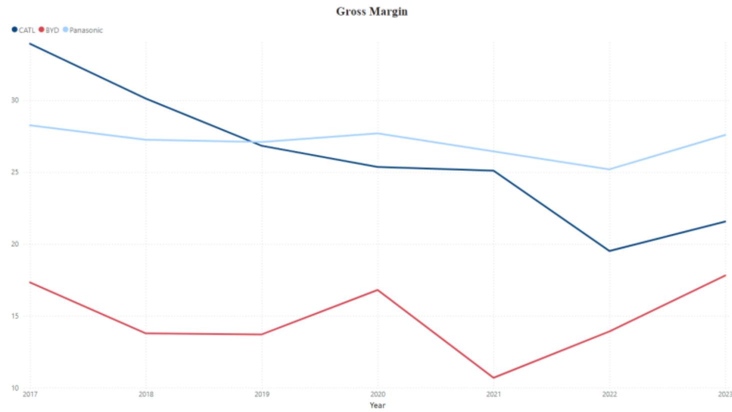

Figure 17: CATL’s gross margin

The revenue of CATL has experienced sharp increases in both 2021 and 2022, following the widespread adoption of electric vehicles and the aggressive push to green transition. The growth rate has slowed down in 2023 primarily due to the large base number and the price pressure from battery overcapacity. The slowdown was attributable to the domestic market, however, CATL’s overseas sales have been soaring as presented in the section above. In essence, battery demands are still on the rise, depicted by the year-on-year growing revenue despite the varying growth rate.

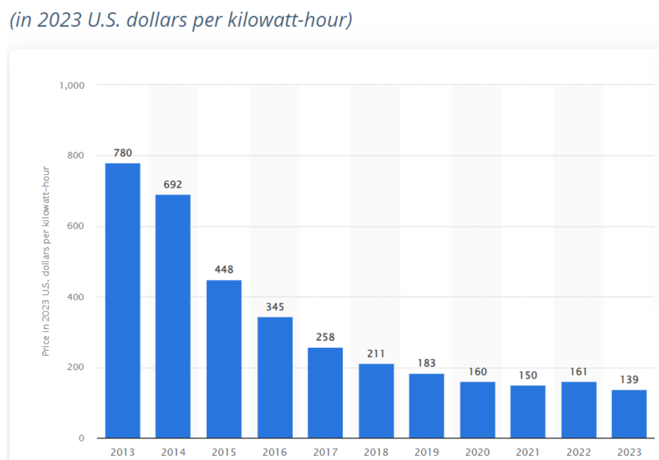

Figure 18: Lithium-ion battery price worldwide from 2013 to 2023

On the contrary, the gross margin has been declining, resulting from the drop in prices of critical materials that drag down the market selling price. As shown in figure 18, the price of battery packs has fallen significantly. In 2023, it showed a slight improvement to 21.56%, ranking second among its rivals, but still inferior to the 33.92% margin in 2017. Nonetheless, its rivals do not show downtrends with similar magnitude as they could be bolstered by their diverse product lines.

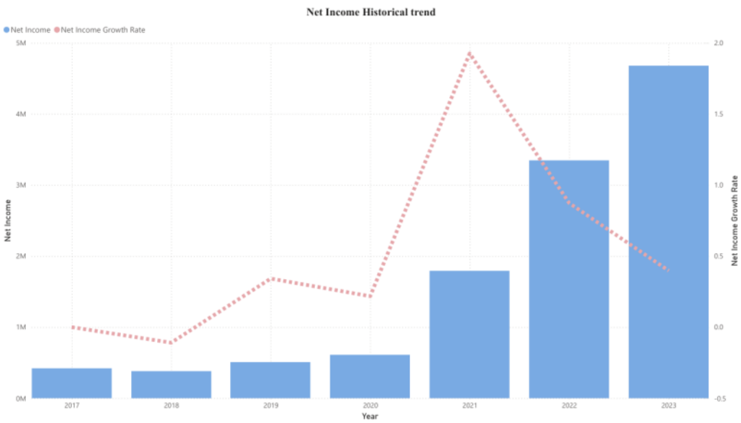

Figure 19: CATL's net income historical trend

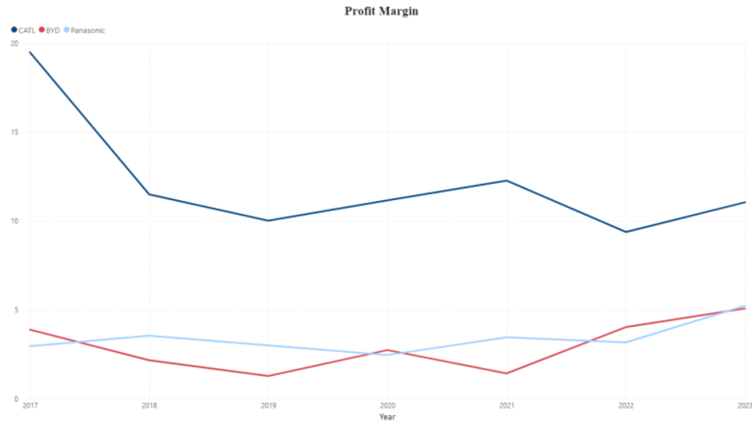

Figure 20: CATL's profit margin

Net income shows a similar trend as revenue. It experienced the same jump in 2021 and demonstrated an overall growing trend. In terms of profit margin, it saw a sharp decline in 2018 but has since stabilized at above 10%, exceedingly outperforming both BYD and Panasonic. This reflects the superior operational management and cost efficiency CATL has achieved from its larger business scale and dominant market share.

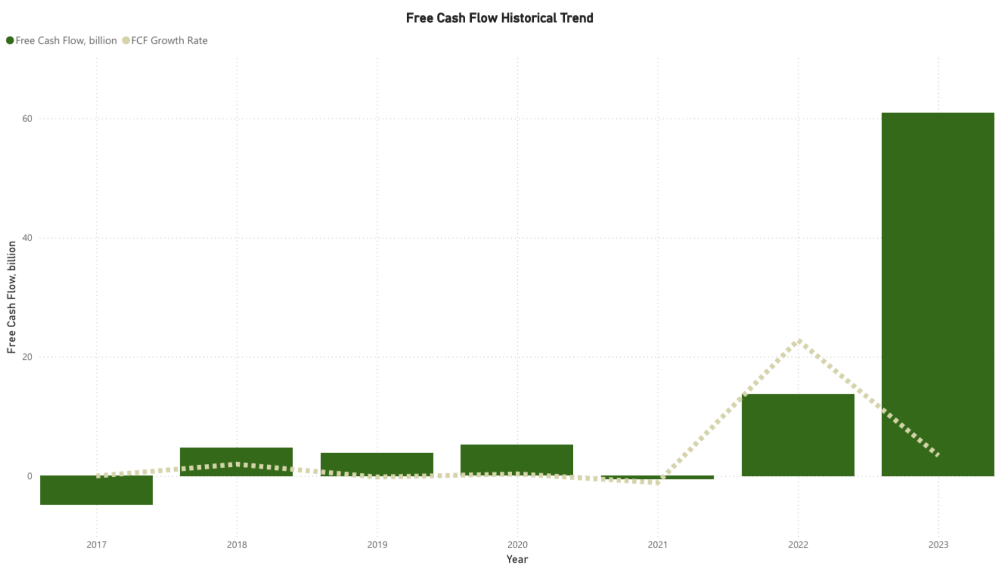

Figure 21: CATL's free cash flow historical trend

A significant free cash flow amounting to CNY 60.91 billion was observed in 2023, making the positive cash flow in previous years seem trivial. However, it is undeniable that CATL has been generating more cash than it used even after accounting for growth investments. The positive free cash flows over the years have contributed to the growing cash amount in the balance sheet, providing ample liquidity and capacity for future ventures and contingency events. While there was a small dip in 2021, it was because of a sudden rise in cash used for investing activities from CNY 14.15 billion to CNY 53.21 billion.

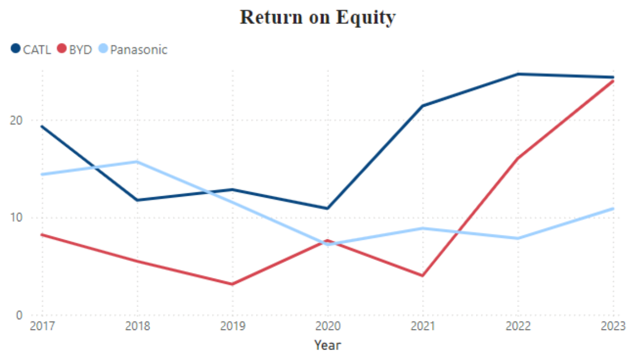

Figure 22: Return on Equity

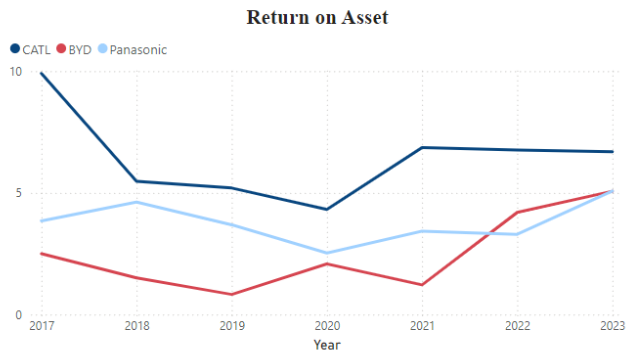

Figure 23: Return on Asset

Return on equity has seen a significant rise over the past few years, resulting from the growing net income. It ranked first among its rivals, at 24.36% ROE. However, BYD almost caught up to CATL, but it could partially be attributable to BYD’s high equity multiplier after applying DuPont analysis. In contrast, Panasonic shows a relatively sluggish performance, hovering between the 10% region. In essence, CATL creates the highest returns for its shareholders.

Additionally, CATL recorded above 5% return on assets over 2021 to 2023, leading its rivals in generating profits from assets. In 2023 alone, CATL achieved 6.69% ROA while BYD and Panasonic touched on 5% for the first time after years of registering under-5-percent ROA. CATL shows better asset management to generate sales, however, it needs to be cautious of the steadily rising performance of BYD, preventing the loss of market share.

Solvency Analysis

Overall, CATL illustrates low-to-moderate reliance on debt as compared to its competitors, with an

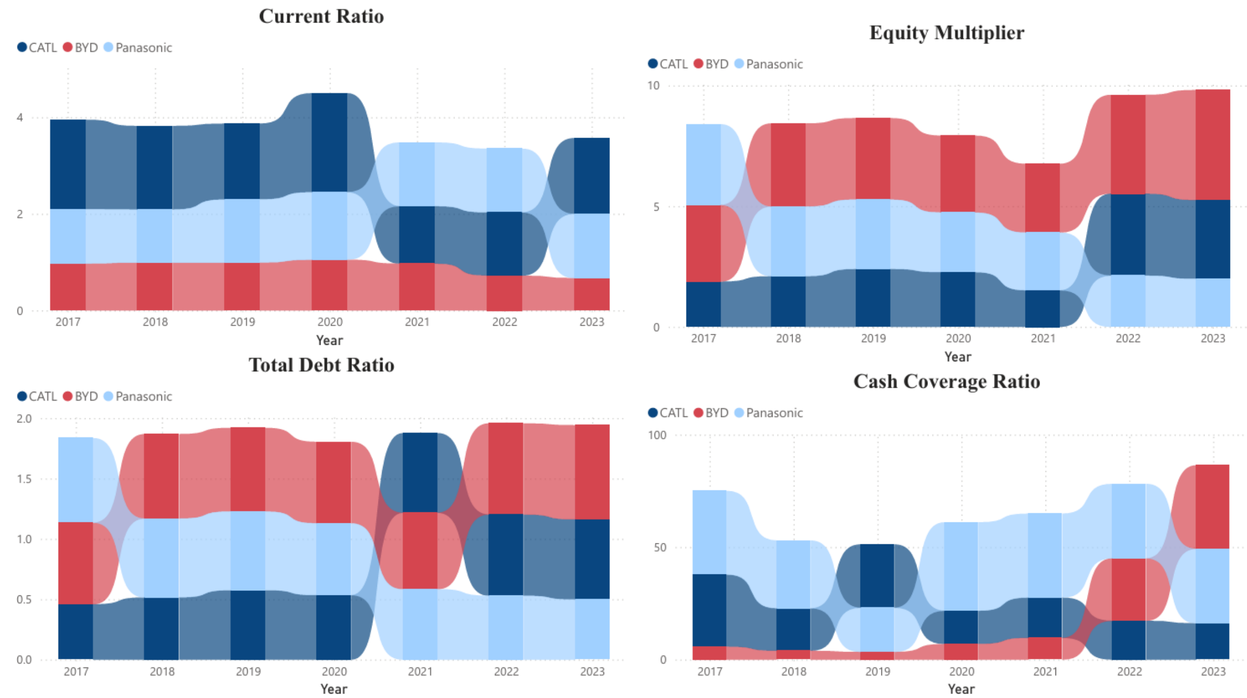

Figure 24: Current ratio, Equity multiplier, Total debt ratio & Cash coverage ratio

average equity multiplier of 2.4 times and total debt ratio of 0.58 over the past seven years. Whereas BYD and Panasonic have an average total debt ratio of 0.7 and 0.6 respectively. In simple terms, three of the companies are using more debt than equity to finance asset purchase, with less than half contributed by the owners or shareholders. Both CATL and BYD show signs of growing dependency on debt recently, following the enormous demand base to serve, while Panasonic is the only one showing a declining trend. Diving deeper, CATL is found to rely more on short-term debts, which demands CATL to maintain a high liquidity position. Not concerningly, CATL has sufficient short-term assets to pay off the current debts, evident by the healthy current ratio of 1.57 in 2023. In short, for every CNY 1 of current debt, there is CNY 1.57 to pay it off.

However, CATL has the lowest cash coverage ratio of 16.15 in 2023 compared to BYD and Panasonic that recorded a ratio of 33.12 and 37.14 respectively, due to its higher debt servicing costs. If an average value over the past seven years was taken, CATL, with a ratio of 20.74, would perform better than BYD that averaged 13.52. Owing to the recent hike in debt servicing costs, the ratio of CATL has been dragged down.

Efficiency Analysis

Individually, CATL has seen an increasing inventory turnover rate since 2020, surpassing Panasonic

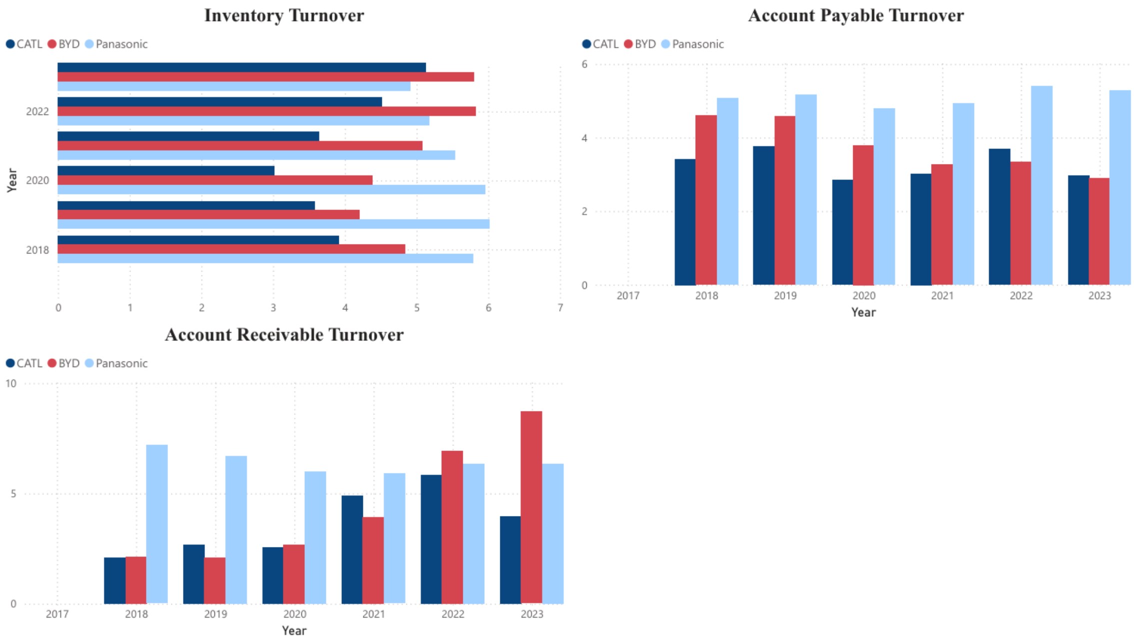

Figure 25: Inventory turnover, Asset receivable turnover & Account payable turnover

in 2023 with a rate of 5.13. Comparatively, CATL lags its rivals. Although being overtaken by CATL in 2023, Panasonic took the lead in early years with almost 6 times inventory turnover before its decline in the past few years. In contrast, BYD has been performing better and better over the years, recording a 5.80 times turnover in 2023.

As seen in the common size analysis, account receivable is one of the largest portions in current assets of CATL with a drastic rise in 2022. A similar composition is observed in BYD. The sharp increase in account receivables could be associated with the soaring battery demand amid the active push to green transition. Following the rise in account receivables, the turnover rate for both companies showed a similar increase, signifying their efforts to maintain a healthy operating cash flows by ramping up debt collection. However, the account receivable turnover rate of CATL dropped in 2023, indicating the possibility of higher credit sales and slow debt collection. At 3.95, it was the lowest compared to BYD’s 8.69 and Panasonic’s 6.34.

Account payable experienced the sharp rise as well, which might concern the investors. The good news is that CATL has managed to sustain its payment to debtors with a consistent turnover rate between 3 to 4 times. Despite the stable turnover rate, CATL lagged behind BYD and Panasonic in the early years as shown in the graph of account payable turnover. Nonetheless, BYD’s high debt financing has resulted in a declining turnover rate from 4.61 in 2018 to 2.89 in 2023, falling behind CATL since 2022. Conversely, Panasonic has demonstrated strong debt-paying ability with a consistent turnover rate above 4, ranking at the top over the past seven years.

Valuation

Referring to Simply Wall Street, CATL is valued at CNY 256.24 per share based on a discounted cash flow model. The company is now priced at CNY 185.15 per share, trading 27.7% below the intrinsic value. Hence, the battery manufacturer is said to be undervalued. Besides, the company recorded the lowest price-to-earnings ratio of 13.86 in the past seven years, signifying a more affordable or attractive pricing for every dollar earned by the company.

From a broader perspective, as discussed in task 1, the China stock market is valued at a huge discount. The current ratio of total market cap to the sum of gross domestic product and total assets of the central bank is 37.72%, way below the 10 years mean of 46.43% (GuruFocus, 2024). Simply said, the China market is considered as moderately undervalued.

Conclusion

In essence, the report issues a BUY Recommendation on Contemporary Amperex Technology with an upside potential of 27.7%. As discussed in the previous sections, CATL holds a promising future in terms of financial performance, industry attractiveness as well as its competitive advantages. CATL grows impressively to be the leader in one of the most important industries, in realizing a successful green transition amid the dire environmental concerns. The battery manufacturer's wide global market share and advanced technological capabilities have established a strong economic moat, making it difficult for competitors to penetrate in the foreseeable future. Even though risks such as trade tension and China market reforms persist, it is believed that its competitive advantages could alleviate potential impacts and outweigh these risks.

Linking back to task 1, the following explanations will reflect the thought process behind for choosing CATL. Amid the restructuring of the Chinese economy, the nation is expected to shift away from traditional industries with low value-added outputs to high value-added industries. As a key player in the green transition, CATL is involved in the highly sophisticated technology industry, solving the pressing global issues. As a result, CATL shows strong product demands as well as the ability to generate positive free cash flow, without neglecting active research and development which has brought consistent innovative breakthroughs that better solve market needs. Furthermore, CATL covers not only the domestic market but also the international arena, expanding its market influence worldwide. As such, the performance of CATL is not solely dependent on one market and would not be severely affected by the current national economic volatility. To sum it up, CATL does not present itself as a value trap.

References

Ambrose, J. (2024, May 8). Renewable energy passes 30% of world's electricity supply. The Guardian. https://www.theguardian.com/environment/article/2024/may/08/renewable-energy-passes-30-of-worlds-electricity-supply

Badlam, J., Cox, J., Kumar, A., Mehta, N., O'Rourke, S. & Silvis, J. (2022). The Inflation Reduction Act: Here's what's in it. McKinsey & Company. https://www.mckinsey.com/industries/public-sector/our-insights/the-inflation-reduction-act-heres-whats-in-it

Best, R. (2023, October 1). China's intervention during the 2015 stock market crash. Investopedia. https://www.investopedia.com/articles/investing/022716/4-consequences-government-intervention-chinas-markets.asp#:~:text=Why%20Did%20China's%20Stock%20Market,to%20limit%20the%20sell%2Doff.

Brunp Recycling. (n.d.). https://en.brunp.com.cn/

CATL. (2023, May 24). CATL launches world's first solar-plus-storage solution with zero auxiliary power supply. https://www.catl.com/en/news/6025.html

CATL. (2024, April 25). CATL unveils Shenxing PLUS, enabling 1,000-km range and 4C superfast charging. https://www.catl.com/en/news/6239.html

Chao, C., Wang, C. & Kam, T. (2024, June 24). Fitch Affirms CATL's Ratings at 'A-', Outlook Stable. Fitch Ratings. https://www.fitchratings.com/research/corporate-finance/fitch-affirms-catl-ratings-at-a-outlook-stable-24-06-2024

Chinese equities – The guide. (2021, April 9). abrdn. https://www.abrdn.com/en-gb/intermediary/insights-and-research/chinese-equity-markets

Deloitte. (2023). Forging ahead on a New Journey Chinese Banking Sector 2022 Review and 2023 Outlook. https://www2.deloitte.com/content/dam/Deloitte/cn/Documents/financial-services/deloitte-cn-fsi-chinese-banking-sector-2022-review-and-2023-outlook-en-231214.pdf

Fitch Ratings. (2023, October 25). CATL's stronger position overseas to offset tougher competition in China. https://www.fitchratings.com/research/corporate-finance/catls-stronger-position-overseas-to-offset-tougher-competition-in-china-25-10-2023

FSMOne. (2023, March 3). Top 10 biggest companies listed in Shanghai Stock Exchange. https://www.fsmone.com.my/funds/research/article-details/268001

Gogoi, D. (2024, May 29). The game of tariffs - US moves to stifle EV battery imports from mainland China. https://www.spglobal.com/mobility/en/research-analysis/briefcase-the-game-of-tariffs-ev-battery-imports.html

GuruFocus. (2024, September 8). Buffett indicator: China stock market valuations and forecasts. https://www.gurufocus.com/global-market-valuation.php?country=CHN

Hang Seng Indexes. (2024, July). Hang Seng Index. https://www.hsi.com.hk/static/uploads/contents/en/dl_centre/factsheets/hsie.pdf

Hawkins, A. (2024, March 18). CATL, the little-known Chinese battery maker that has the US worried. The Guardian. https://www.theguardian.com/world/2024/mar/18/catl-chinese-battery-maker-evs-electric-vehicles

Hilton, I. (2024, March 13). How China became the world's leader on renewable energy. https://e360.yale.edu/features/china-renewable-energy#:~:text=In%202021%2C%20in%20the%20Paris,2030%2C%20and%20to%20generate%2050

International Energy Agency. (2024). Batteries and secure energy transitions. https://iea.blob.core.windows.net/assets/cb39c1bf-d2b3-446d-8c35-aae6b1f3a4a0/BatteriesandSecureEnergyTransitions.pdf

International Energy Agency. (2021, May 5). Comparative life-cycle greenhouse gas emissions of a mid-size BEV and ICE vehicle. https://www.iea.org/data-and-statistics/charts/comparative-life-cycle-greenhouse-gas-emissions-of-a-mid-size-bev-and-ice-vehicle

International Energy Agency. (2024, April). Global EV Outlook 2024: Moving towards increased affordability. https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf

Jarbratt, G., Jautelat, S., Linder, M., Sparre, E., Quan, H. W. & van de Rijt, A. (2023). Enabling renewable energy with battery energy storage systems. McKinsey & Company. https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/enabling-renewable-energy-with-battery-energy-storage-systems

Lam, W. R., Rodlauer, M. & Schipke, A. (2017). Modernizing China: Investing in soft infrastructure. International Monetary Fund.

Lee, M. (2024). China’s road to recovery: how heavily will real estate weigh on growth? Validus. https://www.validusrm.com/2024/04/16/chinas-road-to-recovery-how-heavily-will-real-estate-weigh-on-growth/#:~:text=The%20real%20estate%20sector%20and,growth%20drivers%20for%20the%20country.

Liu, L. (2024, January 29). Property sector critical to growth. China Daily. https://www.chinadaily.com.cn/a/202401/29/WS65b6ff1ba3105f21a507ec78.html

Liu, Y. (2024, June 20). Rolls-Royce partners with Chinese battery maker CATL. China Daily. https://www.chinadaily.com.cn/a/202406/20/WS6673e6cba31095c51c509f62.html

National Bureau of Statistics of China. (2023, January 18). Households' income and consumption expenditure in 2022. https://www.stats.gov.cn/english/PressRelease/202301/t20230118_1892303.html

National Bureau of Statistics of China. (2024, January 18). Households' income and consumption expenditure in 2023. https://www.stats.gov.cn/english/PressRelease/202402/t20240201_1947120.html#:~:text=II.&text=In%202023%2C%20the%20nationwide%20per%20capita%20consumption%20expenditure%20on%20food,the%20per%20capita%20consumption%20expenditure.

Palmer, B. (2023, August 11). Institutional investors vs. retail investors: What's the difference? Investopedia. https://www.investopedia.com/ask/answers/06/institutionalinvestor.asp

PWC. (2023). China banking review 2023. https://www.pwccn.com/en/banking/china-banking-review-2023-embracing-five-major-areas-may2024.pdf

Reuters. (2024, August 29). 2 Republicans ask Pentagon to add Chinese battery maker CATL to restricted list. https://www.scmp.com/news/china/article/3276324/2-republicans-ask-pentagon-add-chinese-battery-maker-catl-restricted-list?module=perpetual_scroll_0&pgtype=article

Simply Wall Street. (n.d.). Contemporary Amperex Technology. https://simplywall.st/stocks/cn/capital-goods/szse-300750/contemporary-amperex-technology-shares

Shepardson, D. (2023, September 27). US lawmakers demand documents on Ford battery partnership with CATL. Reuters. https://www.reuters.com/business/autos-transportation/us-lawmakers-demand-documents-ford-battery-partnership-with-catl-2023-09-27/

Statista. (2023). Distribution of the gross domestic product (GDP) in China in 2023, by industry. https://www.statista.com/statistics/1124008/china-composition-of-gdp-by-industry/#:~:text=In%202023%2C%20the%20industrial%20sector,of%20the%20country's%20economic%20output.

Statista. (2024, August 7). Global market distribution of battery makers for electric vehicles in 2023. https://www.statista.com/statistics/235323/lithium-batteries-top-manufacturers/#statisticContainer

Statista. (2023, December 8). Lithium-ion battery price worldwide from 2013 to 2023. https://www.statista.com/statistics/883118/global-lithium-ion-battery-pack-costs/

Statista. (2021). Share of the global electric vehicles lithium-ion battery manufacturing capacity in 2021 with a forecast for 2025, by country. https://www.statista.com/statistics/1249871/share-of-the-global-lithium-ion-battery-manufacturing-capacity-by-country/

The Star. (2023, November 28). Battery prices are falling again as raw material costs drop globally. https://www.thestar.com.my/business/business-news/2023/11/28/battery-prices-are-falling-again-as-raw-material-costs-drop-globally

Thielen, A. (2024, March 25). EV battery giant CATL unfazed by U.S. restrictions on Chinese companies. Seeking Alpha. https://seekingalpha.com/news/4083379-ev-battery-giant-catl-unfazed-us-restrictions-chinese-companies

United Nations. (n.d.). https://www.un.org/en/climatechange/paris-agreement

United Nations. (2015). Paris Agreement. https://unfccc.int/sites/default/files/english_paris_agreement.pdf

World Economic Forum. (2019). A vision for a sustainable battery value chain in 2030: Unlocking the full potential to power sustainable development and climate change mitigation. https://www3.weforum.org/docs/WEF_A_Vision_for_a_Sustainable_Battery_Value_Chain_in_2030_Report.pdf

Yan, D. (2022, September 29). CATL – The battery giant leading innovation in China A-Shares. https://kraneshares.com/catl-the-battery-giant-leading-innovation-in-china-a-shares/

Zipser, D. (2023). China consumption: Start of a new era. McKinsey & Company. https://www.mckinsey.com/cn/our-insights/our-insights/china-consumption-start-of-a-new-era

Appendix

Appendix 1: CATL's income statement from 2017 to 2023

(in CNY)	2023	2022	2021	2020	2019	2018	2017
Revenue	**400,917,044,900.00**	**328,593,987,500.00**	**130,355,796,400.00**	**50,319,487,697.20**	**45,788,020,642.41**	**29,611,265,434.22**	**19,996,860,806.33**
Cost of goods sold	350,610,617,500.00	293,745,632,000.00	111,367,290,600.00	43,485,843,532.02	38,954,702,306.61	25,729,175,371.38	16,875,255,721.27
Including: Operating costs	309,070,434,000.00	262,049,609,200.00	96,093,722,300.00	36,349,153,592.22	32,482,760,512.62	19,902,284,153.15	12,740,187,148.70
Taxex and surcharges	1,695,507,700.00	907,484,500.00	486,534,200.00	295,129,893.60	272,228,105.62	171,183,911.70	95,900,521.00
Selling expenses	17,954,440,500.00	11,099,401,200.00	4,367,869,400.00	2,216,709,532.73	2,156,553,541.51	1,378,868,425.55	795,766,091.83
Management expenses	8,461,824,300.00	6,978,669,400.00	3,368,937,100.00	1,768,115,240.89	1,832,673,929.87	1,590,659,572.27	1,324,587,823.00
Research & development expenses	18,356,108,400.00	15,510,453,500.00	7,691,427,600.00	3,569,377,694.03	2,992,107,516.52	1,991,000,384.84	1,631,900,455.51
Financial expenses	- 4,927,697,400.00	- 2,799,985,800.00	- 641,200,000.00	- 712,642,421.45	- 781,621,299.53	- 279,733,226.14	42,169,650.35
Including: Interest expenses	3,446,515,800.00	2,132,375,400.00	1,161,100,400.00	640,434,316.54	289,254,465.49	204,435,332.83	98,824,909.51
Interest income	8,321,802,100.00	3,987,365,200.00	2,323,262,000.00	1,494,600,958.67	1,078,256,966.28	565,817,388.30	101,526,949.61
Other income	6,267,388,000.00	3,037,344,800.00	1,673,454,100.00	1,135,940,385.95	646,371,587.96	507,775,215.03	444,421,648.35
Investment income	3,189,201,200.00	2,514,538,600.00	1,232,699,000.00	- 117,648,607.80	- 79,604,902.02	184,397,531.48	1,344,305,303.77
Including： Investment income from associates and joint ventures	3,745,762,000.00	2,614,517,000.00	575,836,900.00	- 4,027,332.09	- 11,899,568.84	- 4,264,014.31	- 49,976,783.37
Gains on Derecognition of Financial Assets Measured at Amortized Cost	- 636,724,600.00	- 530,396,900.00	- 328,331,800.00	- 91,456,141.35	- 100,357,220.91	-	-
Gain on change in fair value	46,270,400.00	400,241,300.00	-	286,915,936.00	27,331,582.10	- 314,247,518.10	-
Credit impairment loss	- 254,041,400.00	- 1,146,247,900.00	- 13,301,600.00	- 341,982,529.63	- 235,676,386.11	-	-
Asset impairment loss	- 5,853,926,900.00	- 2,826,926,600.00	- 2,034,437,800.00	- 827,489,419.04	- 1,434,329,163.69	- 974,912,150.01	- 244,744,030.88
Gain on disposal of assets	16,983,500.00	- 5,322,700.00	- 23,190,300.00	- 9,890,379.23	1,382,204.06	- 91,538,964.57	- 78,311,541.52
Operating profits	**53,718,302,100.00**	**36,821,983,100.00**	**19,823,729,200.00**	**6,959,489,551.43**	**5,758,793,258.10**	**4,168,476,326.68**	**4,832,020,495.66**
Non-operating incomes	503,675,200.00	159,426,800.00	183,039,700.00	94,318,062.61	62,428,112.63	62,303,262.42	18,655,542.93
Non-operating expenses	307,924,000.00	308,553,600.00	119,639,800.00	71,254,204.80	60,456,806.48	25,966,337.17	2,575,814.31
Pretax incomes	53,914,053,300.00	36,672,856,200.00	19,887,129,100.00	6,982,553,409.24	5,760,764,564.25	4,204,813,251.93	4,848,100,224.28
Less: Tax	7,153,018,800.00	3,215,712,700.00	2,026,399,000.00	878,635,356.78	748,090,666.38	468,916,764.21	654,043,646.62
Net income	**46,761,034,500.00**	**33,457,143,500.00**	**17,860,730,100.00**	**6,103,918,052.46**	**5,012,673,897.87**	**3,735,896,487.72**	**4,194,056,577.66**

Appendix 2: CATL’s balance sheet from 2017 to 2023

	2023	2022	2021	2020	2019	2018	2017
Cuurent Asset :							
Cash & Equivalent	264,306,514,700.00	191,043,409,500.00	89,071,889,700.00	68,424,116,053.67	32,269,635,327.07	27,731,189,739.92	14,080,936,626.30
Short term investment	7,767,200.00	1,981,328,100.00	1,363,972,900.00	3,288,071,512.61	1,389,585,592.37	-	-
Derivatives	-	575,638,000.00	243,105,100.00	1,330,347,108.86	1,812,135,529.60	512,661,245.82	13,744,559.02
Total Receivables	65,772,258,000.00	61,492,600,600.00	25,217,376,200.00	21,170,680,072.11	17,988,485,338.20	15,967,748,024.97	12,376,856,841.25
Notes receivable	1,751,724,600.00	3,526,083,700.00	1,463,828,000.00	9,877,156,349.23	9,649,949,692.85	9,742,890,628.44	5,458,335,290.35
Account receivable	64,020,533,400.00	57,966,516,900.00	23,753,548,200.00	11,293,523,722.88	8,338,535,645.35	6,224,857,396.53	6,918,521,550.90
Receivable financing	55,289,318,800.00	18,965,714,600.00	6,486,380,800.00	-	-	-	-
Prepayment	6,962,872,600.00	15,843,284,400.00	6,466,439,300.00	997,118,630.25	538,163,094.42	864,640,798.47	305,835,456.79
Other receivable	3,438,564,000.00	8,678,379,900.00	3,114,909,600.00	3,303,956,813.15	4,568,565,748.80	682,089,431.99	142,941,354.82
Interest receivable	-	-	-	-	-	92,808,117.05	52,310,504.72
Dividend receivable	425,984,800.00	-	-	-	-	-	-
Inventory	45,433,890,100.00	76,668,898,800.00	40,199,691,900.00	13,224,640,950.39	11,480,549,879.88	7,076,101,849.47	3,417,757,092.32
Contract assets	233,964,100.00	174,863,000.00	77,285,500.00	75,269,024.76	-	-	-
Non-current assets that mature within one year	56,827,500.00	403,711,500.00	201,655,900.00	81,548,616.67	-	-	-
Other current assets	8,286,024,600.00	11,907,028,600.00	5,292,231,900.00	969,240,539.21	1,647,816,662.94	1,076,991,664.73	2,696,030,983.77
Total current assets	**449,788,001,700.00**	**387,734,857,000.00**	**177,734,938,700.00**	**112,864,989,321.68**	**71,694,937,173.28**	**53,911,422,755.37**	**33,033,102,914.27**
Non-current assets:							
Avaivable-for-sale financial assets	-	-	-	-	-	1,516,521,098.20	1,961,291,000.00
Long-term receivbles	9,839,600.00	44,316,100.00	619,282,400.00	372,156,591.66	-	-	-
Long-term equity investment	50,027,694,100.00	17,595,207,400.00	10,949,033,600.00	4,813,072,905.14	1,540,452,827.51	965,198,180.81	791,027,220.90
Investment in Other Equity Instruments	14,128,318,200.00	20,491,264,200.00	11,306,853,200.00	1,997,026,992.95	1,530,603,722.80	-	-
Other non-current financial assets	2,816,189,700.00	2,645,306,600.00	1,714,865,300.00	-	-	-	-
Fixed assets	115,387,960,100.00	89,070,834,700.00	41,275,333,300.00	19,621,648,443.02	17,417,348,593.44	11,574,665,757.11	8,219,496,581.74
Construction in progress	25,011,907,100.00	35,397,650,600.00	30,998,159,500.00	5,750,351,820.37	1,996,524,778.01	1,623,838,222.94	2,974,364,031.46
Right-of-use asset	377,934,100.00	849,134,000.00	678,625,300.00	-	-	-	-
Intangible asset	15,675,876,200.00	9,539,963,200.00	4,479,606,400.00	2,517,935,725.46	2,302,317,207.14	1,346,171,137.42	1,408,760,249.33
Goodwill	707,881,700.00	704,065,200.00	527,850,700.00	147,951,887.23	147,951,887.23	100,419,270.78	100,419,270.78
Long-term Prepaid Expenses	4,695,780,000.00	2,294,776,000.00	1,264,339,100.00	363,551,716.95	394,096,018.07	305,828,515.40	139,310,483.53
Deferred tax assets	17,395,585,300.00	9,483,660,400.00	5,542,554,400.00	3,167,109,948.33	2,079,210,533.02	1,240,737,742.63	510,045,198.34
Other non-current assets	21,145,073,300.00	25,101,316,500.00	20,575,419,100.00	5,002,631,587.80	2,248,533,970.82	1,298,901,335.85	525,068,808.10
Total non-current asset	**267,380,039,400.00**	**213,217,494,900.00**	**129,931,922,300.00**	**43,753,437,618.91**	**29,657,039,538.04**	**19,972,281,261.14**	**16,629,782,844.18**
Total asstes	**717,168,041,100.00**	**600,952,351,900.00**	**307,666,860,900.00**	**156,618,426,940.59**	**101,351,976,711.32**	**73,883,704,016.51**	**49,662,885,758.45**
Current Liability :							
Short-term debt	15,181,012,100.00	14,415,402,500.00	12,123,056,900.00	6,335,080,182.17	2,125,646,681.77	1,180,092,100.11	2,245,096,000.70
Transactional financial liabilities	-	-	-	-	286,915,936.00	-	-
Financial liabilities at fair value through profit or loss	-	-	-	-	-	314,247,518.10	-
Derivatives financial liabilities	3,941,409,600.00	-	-	15,636,589,526.90	17,420,197,790.40	-	-
Total Payables	194,553,714,300.00	220,764,444,200.00	107,190,037,600.00	31,271,433,835.42	28,112,335,291.07	18,898,203,153.95	13,790,972,911.63
Advance Receipts	-	-	-	-	6,161,443,242.83	4,994,400,867.91	203,165,478.74
Contract liabilities	23,982,351,900.00	22,444,785,300.00	11,537,915,300.00	6,875,227,795.16	-	-	-
Employee compensation payable	14,846,251,200.00	9,476,018,400.00	5,122,787,800.00	2,657,564,914.42	1,582,275,521.53	1,122,253,456.83	517,308,034.99
Accrued taxes	11,741,826,300.00	4,792,441,200.00	2,403,797,500.00	1,321,059,090.43	962,984,568.04	722,536,564.72	436,196,766.94
Other payables	13,654,001,700.00	15,014,070,600.00	6,176,214,400.00	4,407,776,289.55	5,298,308,992.73	2,924,184,174.56	332,362,146.18
Including: Interest payables	-	-	-	-	-	19,842,845.23	15,205,262.28
Dividend payable	29,915,700.00	8,319,600.00	6,590,800.00	6,172,824.12	2,418,687.96	-	-
Proceeds from Underwriting Securities as Agent	-	-	-	1,349,038,696.49	1,077,468,496.09	-	-
Held-for-sale liabilities	-	-	-	760,008,999.58	-	-	-
Non-current laibilities maturing within one year	7,008,874,200.00	7,232,224,400.00	3,548,532,800.00	1,349,038,696.49	1,077,468,496.09	929,024,032.37	364,944,599.97
Other current liabilites	2,091,627,800.00	1,622,032,600.00	1,242,490,300.00	760,008,999.58			
Total current liabilities	**287,001,069,100.00**	**295,761,419,300.00**	**149,344,832,600.00**	**54,977,189,803.22**	**45,607,378,729.06**	**31,084,941,868.55**	**17,890,045,939.15**
Non-current liability:							
Long-term debt	83,448,981,700.00	59,099,358,400.00	22,119,078,800.00	6,068,163,254.20	4,980,563,181.26	3,490,767,815.96	2,129,095,275.13
Bonds payable	19,237,014,400.00	19,177,888,600.00	15,855,052,000.00	14,382,255,950.87	1,508,339,196.70	-	-
Lease liabilities	283,296,200.00	572,350,200.00	395,304,000.00	-	-	-	-
Long-term payable	1,520,256,200.00	1,050,000,000.00	1,010,000,000.00	1,193,938,630.30	873,618,580.61	943,414,523.31	896,045,177.66
Provisions (for Liabilities)	51,638,913,500.00	19,697,374,600.00	9,953,762,700.00	6,797,704,877.32	5,289,773,262.40	2,512,382,681.52	1,789,007,284.61
Deferred earnings	21,448,987,400.00	19,966,701,700.00	12,099,359,100.00	3,918,939,197.71	813,236,654.86	611,042,047.22	419,460,871.47
Deferred tax liabilities	1,364,905,800.00	1,807,813,000.00	1,038,576,900.00	85,518,810.08	91,191,949.71	40,984,489.33	68,992,112.81
Other non-current liabilities	31,341,466,200.00	6,910,284,100.00	3,228,720,400.00	-	-	-	-
Total non-current liabilities	**210,283,821,400.00**	**128,281,770,600.00**	**65,699,853,800.00**	**32,446,520,720.48**	**13,556,722,824.54**	**7,598,591,557.34**	**5,301,600,721.68**
Total liabilities	**497,284,890,500.00**	**424,043,189,900.00**	**215,044,686,400.00**	**87,423,710,523.70**	**59,164,101,553.60**	**38,683,533,425.89**	**23,191,646,660.83**
Owner's equity:							
Share capital	4,399,041,200.00	2,442,514,500.00	2,330,851,200.00	2,329,474,028.00	2,208,399,700.00	2,195,017,400.00	1,955,193,267.00
Capital reserve	87,907,212,600.00	88,904,372,100.00	43,163,696,500.00	41,662,151,603.08	21,630,448,577.59	21,372,918,712.25	15,354,587,816.94
Less： Treasury stock	1,572,971,600.00	253,991,000.00	443,534,900.00	710,020,552.82	1,074,894,790.00	793,701,060.00	-
Other comprehensive income	1,528,223,100.00	8,931,300,000.00	4,208,319,800.00	1,126,992,951.02	620,819,644.93 -	336,839,207.93	248,500,011.58
Reserves	9,355,000.00	-	-	-	-	-	-
Surplus reserve	2,192,566,200.00	1,214,302,900.00	1,158,471,200.00	1,157,782,633.55	1,097,245,469.55	985,878,418.69	638,253,676.69
Undustributed profits	103,244,625,900.00	63,242,753,100.00	34,095,467,500.00	18,640,918,703.75	13,652,965,292.41	9,515,006,632.30	6,504,904,798.90
Total equity attributable to shareholders	197,708,052,400.00	164,481,251,600.00	84,513,271,300.00	64,207,299,366.58	38,134,983,894.48	32,938,280,895.31	24,701,439,571.11
Minority interest	22,175,098,200.00	12,427,910,400.00	8,108,903,200.00	4,987,417,050.31	4,052,891,263.24	2,261,889,695.31	1,769,799,526.51
Total owner's equity	**219,883,150,600.00**	**176,909,162,000.00**	**92,622,174,500.00**	**69,194,716,416.89**	**42,187,875,157.72**	**35,200,170,590.62**	**26,471,239,097.62**
Total liabilities and owner's equity	**717,168,041,100.00**	**600,952,351,900.00**	**307,666,860,900.00**	**156,618,426,940.59**	**101,351,976,711.32**	**73,883,704,016.51**	**49,662,885,758.45**

Appendix 3: CATL’s cash flow statement from 2017 to 2023

	2023	2022	2021	2020	2019	2018	2017
Cash flow from operating activities:							
Net income	417,943,222,600.00	305,775,248,400.00	130,616,575,600.00	54,002,988,575.54	52,149,078,427.56	33,853,639,616.99	18,872,908,615.92
Tax refunds	12,739,610,500.00	9,478,689,900.00	414,542,800.00	151,928,522.94	395,686,584.53	37,127,687.41	21,013,193.89
Other operating activities	15,724,664,000.00	14,557,214,600.00	14,298,549,300.00	6,397,335,867.06	3,763,244,302.06	2,274,383,357.82	637,638,992.58
Total cash inflows	446,407,497,100.00	329,811,152,800.00	145,329,667,700.00	60,552,252,965.54	56,308,009,314.15	36,165,150,662.22	19,531,560,802.39
Cash Paid for Goods Purchased and Services Received	310,521,178,200.00	235,327,104,000.00	86,393,876,300.00	33,461,951,736.87	33,260,080,970.87	19,041,614,626.94	12,357,592,298.17
Employee compensation	21,140,596,600.00	18,157,352,000.00	9,423,183,000.00	4,027,848,128.71	3,704,062,487.97	2,299,961,469.07	2,122,637,417.20
Taxes	17,117,191,500.00	10,529,733,400.00	4,127,789,000.00	2,265,587,429.63	2,688,800,053.78	1,868,081,717.00	1,493,296,692.69
Cash outflow for other operating-related activities	4,802,406,300.00	4,588,120,100.00	2,476,810,700.00	2,366,963,038.37	3,183,111,244.73	1,639,227,148.68	1,108,823,984.08
Total cash outflows	353,581,372,700.00	268,602,309,600.00	102,421,659,000.00	42,122,350,333.58	42,836,054,757.35	24,848,884,961.69	17,082,350,392.14
Net cash flow from operating activities	**92,826,124,400.00**	**61,208,843,300.00**	**42,908,008,700.00**	**18,429,902,631.96**	**13,471,954,556.80**	**11,316,265,700.53**	**2,449,210,410.25**
Cash flow from investing activities:	-	-	-				
Cash Received from Investment Withdrawals	7,651,158,400.00	1,307,996,200.00	571,745,100.00	44,779,761.91	0.23	5,258,600.00	448,002,008.14
Investment Income	1,711,392,900.00	740,372,300.00	228,638,800.00	24,087,086.33	2,563,085.15	3,655,688.20	2,565,089.09
Disposal of fixed assets, intangible assets and other long-term assets	12,853,500.00	593,900.00	3,223,100.00	24,649.08	15,088,255.27	294,506.10	572,921.94
Disposal of subsidiaries and other business units	3,306,900.00	-	58,414,400.00	-	10,026,813.18	-	-
Cash inflow from other investment-related activities	1,239,798,700.00	1,531,307,500.00	4,094,314,000.00	2,735,874,410.58	15,477,402,748.49	70,629,509.16	650,369,140.52
Total cash inflow from investing activities	10,618,510,400.00	3,580,269,900.00	4,956,335,300.00	2,804,765,907.90	15,505,080,902.32	79,838,303.46	1,101,509,159.69
Purchase of fixed assets, intangible asset and other long term assets	33,624,896,500.00	48,215,268,100.00	43,767,770,800.00	13,302,355,759.04	9,626,986,411.07	6,629,274,672.98	7,180,281,117.24
Cash paid for investment	5,649,689,400.00	12,764,660,700.00	11,725,718,900.00	4,088,418,479.98	907,758,218.55	192,025,942.30	1,665,503,314.78
Net Cash Received from Subsidiaries and Other Operating Units	321,445,400.00	-	295,800,300.00	-	-	-	-
Cash outflow for other investing-related activities	210,243,200.00	6,740,182,400.00	2,948,104,700.00	466,447,835.43	3,114,021,020.85	12,746,210,861.04	656.00
Total cash outflows from investinng activities	39,806,274,600.00	67,720,111,200.00	58,737,394,700.00	17,857,222,074.45	13,648,765,650.47	19,567,511,476.32	8,845,785,088.02
Net cash flow from investing activities	**- 29,187,764,200.00**	**- 64,139,841,300.00**	**- 53,781,059,400.00**	**- 15,052,456,166.55**	**1,856,315,251.85**	**- 19,487,673,172.86**	**- 7,744,275,928.33**
Cash flow from financing activities:	-	-	-				
Cash Received from Investment Contributions	3,323,996,400.00	47,455,207,300.00	1,551,412,800.00	20,536,360,992.16	1,217,784,889.68	6,274,955,935.37	6,178,645,978.00
Including: Cash Received from Minority Shareholders' Investments in Subsidiaries	2,926,448,200.00	2,092,259,300.00	795,200,000.00	923,523,424.41	721,974,398.68	108,785,163.60	1,168,645,100.00
Issuance of debt	46,595,746,400.00	50,957,726,600.00	26,276,582,500.00	9,450,920,678.78	4,616,632,167.94	4,123,196,326.07	4,476,769,278.83
Cash Received from Other Financing Activities	366,758,000.00	5,208,177,600.00	3,235,028,400.00	13,199,266,407.50	1,498,800,000.00	510,282,473.44	9,028,894.69
Total cash inflow from financing activities	50,286,500,800.00	103,621,111,500.00	31,063,023,600.00	43,186,548,078.44	7,333,217,057.62	10,908,434,734.88	10,664,444,151.52
Retirement of debt	23,795,322,000.00	17,605,770,500.00	5,457,907,900.00	4,743,701,233.75	2,418,452,429.42	3,493,721,841.00	1,421,628,377.86
Total cash dividends paid	9,481,092,900.00	3,551,469,400.00	1,568,025,100.00	898,809,819.12	573,302,388.77	216,119,536.13	81,600,198.58
Including: divident payment to subsidiaries and minority shareholders	469,828,200.00	-	-	7,825,842.99	-	-	-
Cash paid for Other Financing Activities	2,293,723,300.00	197,440,400.00	378,512,600.00	112,602,486.19	173,132,704.46	155,877,429.51	228,481,941.80
Total cash outflow from financing activities	35,570,138,100.00	21,354,680,400.00	7,404,445,600.00	5,755,113,539.06	3,164,887,522.65	3,865,718,806.64	1,731,710,518.24
Net cash flow from financing activities	**14,716,362,700.00**	**82,266,431,200.00**	**23,658,578,000.00**	**37,431,434,539.38**	**4,168,329,534.97**	**7,042,715,928.24**	**8,932,733,633.28**
Impact of exchange rate	2,181,446,800.00	2,788,148,900.00	- 711,778,200.00	- 576,950,661.91	14,801,651.37	27,440,781.07	- 13,763,656.75
Net increase in cash & equivalent	**80,536,169,700.00**	**82,123,582,000.00**	**12,073,749,200.00**	**40,231,930,342.88**	**19,511,400,994.99**	**- 1,101,250,763.02**	**3,623,904,458.45**
Opening balance of cash & equivalent	**157,629,317,200.00**	**75,505,735,200.00**	**63,431,986,000.00**	**23,200,055,644.02**	**3,688,654,649.03**	**4,789,905,412.05**	**1,166,000,953.60**
Balane of cash & equivalent at the end of the period	**238,165,486,900.00**	**157,629,317,200.00**	**75,505,735,200.00**	**63,431,985,986.90**	**23,200,055,644.02**	**3,688,654,649.03**	**4,789,905,412.05**